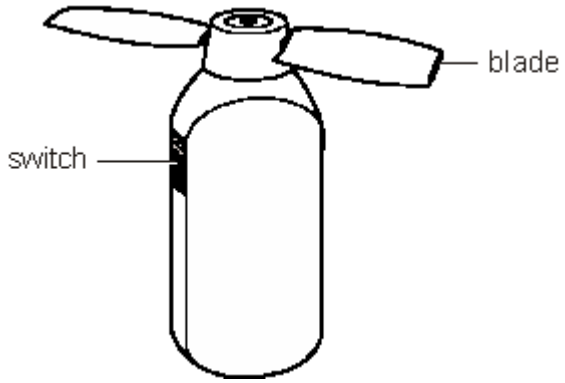
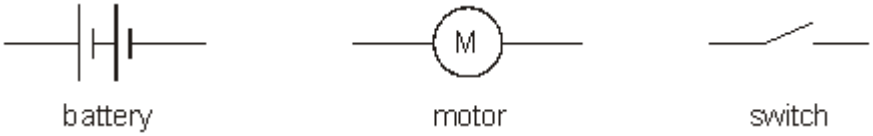
Q1. Susan has a small fan to keep herself cool.
When she switches it on, a motor turns the blades to blow air.



(a) The diagrams below show the symbols for a battery, a motor and a switch.



In the space below, draw a series circuit diagram for the fan using these symbols.

1 mark

(b) (i) Which part provides energy for the circuit?

.....

1 mark

(ii) Some of this energy is used to turn the blades.
The rest of the energy is wasted.

Complete the sentence below. Choose words from the list.

chemical **heat** **light** **sound**

1 mark

When the blades are turning, energy is wasted as

..... energy and ..... energy.

1 mark

(c) Susan built a circuit using a battery, a motor and a switch.
She closed the switch to turn the motor on.

(i) Susan added a bulb to the circuit.
The current in the circuit **decreased.**

How did this affect the motor?

.....

1 mark

(ii) Susan removed the motor from the circuit.
The current in the circuit **increased**.

How did this affect the bulb?

.....

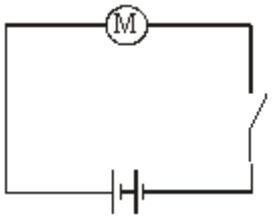
1 mark

maximum 6 marks

Mark schemes

Q1.

(a) •



accept a battery drawn as  *,*

***all three** symbols must be connected in series*

***all three** symbols must be drawn correctly*

do not accept a switch drawn as 

do not accept a battery drawn as 

do not accept a motor drawn as 

1 (L4)

(b) (i) • the battery or cell(s)

accept 'batteries'

1 (L4)

(ii) • heat

1 (L4)

• sound

accept 'thermal'

answers may be in either order

1 (L4)

(c) (i) • it turned more slowly

accept 'not as much energy'

accept 'it was less powerful'

accept 'it does not work as well'

accept 'it would stop' or 'it would not go round'

'it heats up' is insufficient

1 (L4)

(ii) • it was brighter

accept 'blew it' ***or*** *'it went out'*

accept 'it became hotter' ***or*** *'it heated up'*

1 (L4)

[6]